# IMU-Based Wrist Rotation Control of a Transradial Myoelectric Prosthesis

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***Abstract*— This paper describes a control method intended to facilitate improved control of a myoelectric prosthesis containing a wrist rotator. Rather than exclusively utilizing electromyogram (EMG) for the control of all myoelectric components (e.g., a hand and a wrist), the proposed controller utilizes inertial measurement (from six-axis inertial measurement unit (IMU)) to sense upper arm abduction/adduction, and uses this input to command a wrist rotation velocity. As such, the controller essentially substitutes shoulder abduction/adduction in place of agonist/antagonist EMG to control wrist angular velocity, which preserves EMG for control of the hand (or other arm components). As a preliminary assessment of efficacy, the control method was implemented on a transradial prosthesis prototype with a powered wrist rotator and hand, and experimentally assessed on five able-bodied subjects who wore the prototype using an able-bodied adaptor and one transradial amputee subject while performing assessments representative of activities of daily living. The assessments compared the (timed) performance of the combined EMG/IMU-based control method with a (conventional) sequential EMG control approach. Results of the assessment indicate that the able-bodied subjects were able to perform the tasks 33% faster on average with the EMG/IMU-based method, relative to a conventional sequential EMG method. The same assessment was subsequently conducted using a single transradial amputee subject, which resulted in similar performance trends, although with a somewhat lessened effect size—specifically, the amputee subject was on average 22% faster in performing tasks with the IMU-based controller.**

***Index Terms*— Artificial limbs, medical robotics, prosthetics.**

## I. Introduction

SURVEYS of upper extremity amputees indicate a desire for improved functionality and more intuitive, effective, and coordinated control of their arm prostheses [1]–[3]. Recent studies suggest wrist functionality may offer substantial functional benefit to upper extremity amputees, and in particular that inclusion of wrists may substantially reduce the extent of compensatory movements required to perform activities of daily living (ADLs) [4], [5]. A number of myoelectric wrist joints have been developed, several of which are commercially available for use [6]. Despite this, as has been recognized in prior usage studies, user acceptance of additional degrees of freedom (DOFs) requires that the additional DOFs be easily selectable, entail minimal (or negligible) latency, and be speed-controllable [7]. The current method of control for commercially-available myoelectric wrists is a two-site sequential electromyogram (EMG) control interface, where a single pair of EMG electrodes are used to control a single myoelectric DOF at any given time, and where a brief co-contraction of the EMG sites is used to select (via multiplexing) which myoelectric component is actively controlled by the EMG input at any given time. The actively controlled component is typically controlled such that the movement velocity of the component is proportional to the EMG amplitude associated with a given movement direction. As such, although wrist DOFs provide functional value in able-bodied subjects [4], in the case of sequential EMG control (i.e., operation of a single DOF at a time), the value of additional wrist DOFs is less clear, particularly since the sequential method becomes increasingly inefficient with additional myoelectric DOFs in the arm. As such, the functional value of wrist DOFs in a myoelectric prosthesis is dependent on the efficacy of the control method used to control them.

In order to facilitate wrist control, methods other than sequential control have been proposed and demonstrated, such as those described in [8]–[11]. These methods include EMG pattern recognition approaches, and simultaneous proportional myoelectric control approaches, which provide simultaneous control of hand/wrist movements for transradial prostheses. Such methods increase the amount of control information by utilizing multiple EMG sites (i.e., rather than two), which offers enhanced ability to control multi-component myoelectric prostheses, but also increases the complexity of the electrode interface. Regardless, the ability of such methods to offer improved control remains limited by the number of accessible EMG channels.

In an effort to expand the available information from the user with which to control a multifunction myoelectric arm prosthesis, the authors propose herein to supplement EMG information with inertial measurements provided by recently available multi-axis low-cost semiconductor inertial measurement units (IMUs), which can provide real-time measurement of residual limb motion, and to use knowledge of that motion to enhance control of the prosthesis. The use of IMU information to improve arm control was also suggested

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in a relatively recent perspectives article [12], and the idea of leveraging measurement of residual limb motion to enhance prosthesis control was additionally the topic of a recent paper investigating such synergies [13]. Like EMG, movement of the residual limb provides a measure of volitional intent.

In an effort to leverage residual limb motion as input, the authors propose herein a controller that utilizes an IMU on the prosthesis to measure angular rotation of the forearm about the wrist rotation axis, resulting presumably from shoulder motion, and incorporates this input to further drive wrist rotation. The approach is based loosely on the existence of movement synergies between the shoulder and wrist joints, such as those described in [14]–[17]. In order to provide a preliminary assessment of the efficacy of the proposed control approach, the authors conducted assessments on five able-bodied subjects using a myoelectric transradial prosthesis prototype (via an arm adaptor), and subsequently on a transradial amputee subject, in which the ability of subjects to complete manipulation tasks using the IMU-based wrist controller was compared to their ability to do so using a conventional sequential EMG control approach.

## II. IMU-BASED CONTROL APPROACH

Individuals with upper extremity amputation generally compensate for the lack of wrist motion control during manipulation tasks with compensatory motions of their torso and upper arm [15], [16]. Observations of these motions indicate that individuals use a high degree of shoulder ab/adduction to accommodate wrist mobility deficiencies in rotation (i.e., pronation/supination), and when the shoulder range of motion is exceeded, shoulder ab/adduction is supplemented by movement of the torso in the frontal plane (i.e., sideways leaning). Such observations are supported by studies reported in [13], which give quantitative evidence for the coupling between shoulder ab/adduction and wrist pronation/supination in manipulation tasks. These motion synergies provided inspiration for the controller described here, specifically with regard to coordinating the motion between the intact shoulder and prosthetic wrist, thereby facilitating control of the wrist joint, and reducing the need for compensatory movements of the torso.

As previously mentioned, synergies between shoulder movement and wrist rotation were modeled in biomechanical studies on able-bodied subjects [14], [17]–[19]. The models established in these studies are essentially postural relationships, in which the wrist joint angle is related to the shoulder joint angle. Strictly employing such a relationship presents two problems for purposes of real-time control of a prosthesis. First, parameterizations of the relationships are generally dependent on the starting and ending points of a given movement. Since one cannot in general know the endpoint of the movement during the movement, these model parameters are not available in real time. Second, the joint synergy models were created in the context of reaching movements, and although the arm frequently performs reaching tasks, it also performs a variety of other activities, and as such, an invariant relationship between these joints would likely interfere with other arm functions.

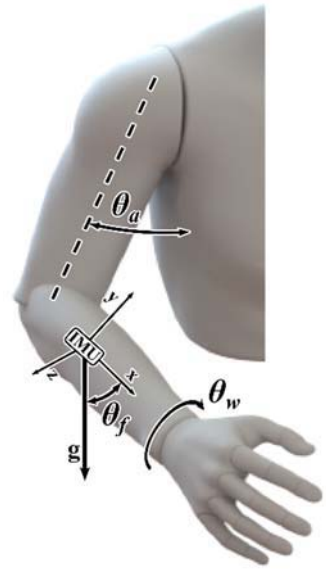


Fig. 1. Diagram showing location of IMU on arm and motions used in the IMU-based controller. $x, y, z$: local coordinate frame of the IMU $\theta_a$: shoulder abduction angle $\theta_w$: wrist pronation/supination angle $\theta_f$: forearm elevation angle (defined with respect to gravitDiagram showing location of IMU on arm and motions used in the IMU-basey vector).

As such, rather than incorporate these models in the postural forms presented in the original respective studies, the authors have generalized and abstracted the kinematic synergy between shoulder ab/adduction and wrist rotation to relate shoulder ab/adduction to wrist rotation (i.e., pronation/supination) angular velocity. Such a relationship enables non-single-valued mappings between shoulder angle and wrist angle in a straightforward manner. Specifically, wrist angle is generally a multi-valued function of shoulder angle. For example, when picking up an object on a table, for a given shoulder orientation, the hand might be oriented with the palm in the horizontal plane (i.e., with the wrist pronated), or in the vertical plane (i.e., with the wrist supinated), depending on the orientation of the object to be grasped. A relationship between joint angles would need to be non-single-valued in order to provide such behavior. Although such a relationship could be constructed for use in a real-time control system, the use of a "velocity control" approach as employed here enables non-single-valued behavior in a straightforward implementation, while still maintaining the essence of the movement synergy between shoulder ab/adduction and wrist pronation/supination.

Although the input is conceptually the shoulder ab/adduction angle, the controller approximates this measure by measuring the upper arm angle with respect to the vertical in the frontal plane, ($\theta_a$ in Fig. 1), which can be measured with an inertial measurement unit (IMU) mounted on the prosthesis. This angle will be the same as the shoulder ab/adduction angle in cases when the torso remains vertical. When the controller is active, the wrist (rotational) angular velocity is directly proportional to $\theta_a$, as measured from a neutral position where the arm is slightly abducted away from the torso, but where the plane defined by the forearm and upper arm remains essentially parallel to the midsagittal plane. A positive value of $\theta_a$ (upper arm angle), which

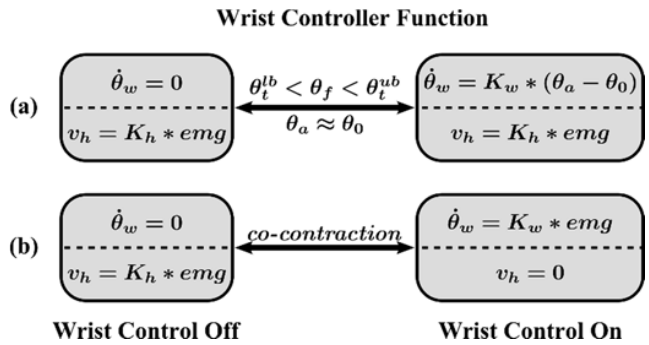


Fig. 2. Wrist controller configurations used in assessment. (a) IMU-based wrist controller, EMG controlled hand. (b) EMG sequential controller, EMG controlled wrist and hand.

indicates shoulder abduction, results in a wrist angular velocity in the pronation direction, whereas a negative value (shoulder adduction) will generate a wrist angular velocity in the supination direction. Returning the upper arm to a small band around the neutral position (i.e., $\theta_0^{lb} \leq \theta_a \leq \theta_0^{ub}$) will result in zero angular velocity (i.e., will maintain a stationary wrist angle). The coupling between upper arm angle and wrist rotation velocity ($\dot{\theta}_w$) is adjusted by specifying the velocity gains $k_{ad}$ and $k_{ab}$ for the adduction and abduction regions, respectively. Referring to the angles defined in Fig. 1, the wrist angular velocity when in the IMU-based control mode is as follows:

$$\dot{\theta}_w = \begin{cases} k_{ad}\left(\theta_a - \theta_0^{lb}\right), & if\ \theta_a < \theta_0^{lb} \\ 0, & if\ \theta_0^{lb} \leq \theta_a \leq \theta_0^{ub} \\ k_{ab}\left(\theta_a - \theta_0^{ub}\right), & if\ \theta_a > \theta_0^{ub} \end{cases} \quad (1)$$

The user activates or deactivates the wrist controller (1) via a compound switching condition. Specifically, toggling into and out of (1) requires the upper arm be held within the neutral position band ($\theta_0^{lb} \leq \theta_a \leq \theta_0^{ub}$), and also that the elbow be flexed so that the forearm elevation angle, as measured by the IMU and defined as $\theta_f$, falls within a band specified by $\theta_t^{lb}$ and $\theta_t^{ub}$ (which corresponds to holding the forearm in an approximately vertical orientation). When these conditions are satisfied for a brief period (e.g., 0.5 s), the IMU-based control of the wrist mode is toggled on or off. Fig. 2 shows a state chart illustrating the switching conditions of the IMU-based wrist controller, as compared to a traditional sequential EMG co-contraction controller.

It is important to note that the IMU-based wrist controller does not require the use of EMG (for control of the wrist joint), and as such, EMG is preserved for other use in the arm controller. In the case of a transradial prosthesis (such as the one used herein), EMG can be used exclusively for control of a myoelectric hand, which enables simultaneous control of the hand and wrist by using simultaneous EMG and upper arm movement, respectively.

## III. Assessment Procedure

The ability of the IMU-based wrist controller to provide control of wrist rotation during manipulation tasks was assessed and compared with a conventional sequential EMG control approach in a set of experiments with five able-bodied subjects and subsequently with one transradial amputee subject. In the case of the able-bodied subjects, an adaptor was employed so that the subjects could affix the prosthesis prototype to their intact arm. In the case of the amputee subject, the prosthesis prototype was mounted directed to his daily-use socket via a standard myoelectric quick-connect interface. Details regarding the implementation of the controller are described in the following sections.

### A. Prosthesis Prototype

The controller was implemented in a transradial arm prosthesis prototype consisting of a multigrasp hand and single DOF wrist rotator. The multigrasp hand used in the assessments was previously described in [20] and previously demonstrated in functional assessments in [21], and the wrist is described in [22]. The hand is capable of performing eight different grasps and postures, which are accessed via a multigrasp myoelectric controller described in [23]. The hand controller enables access to the set of grasps and postures provided by the hand via a finite state machine that uses a (standard) two-site EMG input in addition to measurement of the hand configuration and joint torques.

The wrist rotator used in the assessments was developed by the authors in order to enable the implementation and assessment of the IMU-based wrist control approach described here. The wrist rotator unit consists of a single DOF for pronation/supination, and provides an experimentally measured active torque of 1.5 Nm, maximum rotational speed of 145 deg/s, and maximum range of motion of ±180 deg, as configured for the experiments described here. The wrist is non-backdrivable, and therefore provides a holding torque considerably greater than the 1.5 Nm of active torque. The prosthesis prototype was designed for operation with an onboard microcontroller and battery, but for the purposes of the experiments reported here, the hand and wrist were tethered to a benchtop DC power supply and a PC running Simulink Real-Time Workshop, which provided high-level arm control (i.e., equation (1)) and facilitated data acquisition. The high-level PC control was supplemented by an embedded system on the arm, which performed IMU and EMG signal conditioning and low-level position and current control and motor commutation via a 50 MHz Microchip dsPIC33 and an 80 MHz Microchip PIC32 microcontroller.

### B. Controller Parameters

The control law described by (1) was implemented on the prosthesis prototype with the nominal controller parameters listed in Table I (although it should be noted that one of the subjects preferred thresholds that were 10-15 degrees offset from these values). In all cases, the thresholds were adjusted (as desired by each subject) during initial practice sessions, and were kept constant throughout all subsequent assessments. Note that the deadband for $\theta_a$ is slightly offset in the abduction direction, which made it easier for subjects to perform the adduction movements required for wrist supination. Similarly, the trigger band for $\theta_f$ is slightly less than 90° , which compensates for the slight angle in the elbow when it is fully flexed. The gains that determine the speed of rotation

TABLE I
IMU-BASED CONTROLLER PARAMETERS

| Parameter | | Value |
|---|---|---|
| $\theta_a$ | $\theta_0^{ub}$ | 25° |
| | $\theta_0^{lb}$ | -10° |
| $\theta_f$ | $\theta_t^{ub}$ | 170° |
| | $\theta_t^{lb}$ | 145° |
| $k$ | $k_{ab}$ | 1.5 |
| | $k_{ad}$ | 1.5 |

($k_{ad}$ and $k_{ab}$) were determined by trial and error until the subjects were satisfied with the trade-off between speed of response and ease of control. It is possible that increasing the gains could increase controller performance, but only if the user is comfortable with the sensitivity of control.

### C. Able-Bodied Adaptor

Able-bodied subjects wore the prosthesis prototype using an able-bodied adapter, which allowed free elbow motion while immobilizing the intact wrist. The combined mass of the able-bodied adapter and transradial prosthesis prototype was approximately 1.5 kg (or 700g without the adaptor).

### D. Motion Capture System

In addition to the adaptor and prosthesis, the subjects also donned a motion capture shirt on their upper body, which used markers to track the location of their lower back, neck, upper arm, and wrist. Motion capture information was used to compare the compensatory motions of the subjects between the different controllers while performing the assessments. The motion capture for the able-bodied subjects was performed using twelve OptiTrack S250e high-speed infrared cameras running at 120 Hz using ARENA motion capture software. Motion of the upper body was measured using four rigid bodies consisting of 3-4 markers each. The four markers were attached to the lower back (aligned with the spine and situated approximately level with the hips), the upper back (aligned with the spine at the base of the neck), the upper arm (middle of the arm and oriented parallel to the midsagittal plane), and the hand (placed on the posterior aspect of the hand). These positions were chosen to give relative orientation information between all of the major links in the kinematic chain involved in the tasks: the back, the upper arm, and the forearm. A subject wearing the able-bodied adapter and motion capture markers is shown in Fig. 3.

### E. Obtaining Arm Angles from IMU Measurements

The arm orientation information used by the controller was computed in real-time with data measured by a six-axis IMU mounted to the forearm of the prosthesis prototype. Fig. 1 illustrates the placement of the IMU on the forearm, showing the orientation of the local frame and the directions of upper arm frontal plane angle ($\theta_a$), wrist pronation ($\theta_w$), and forearm

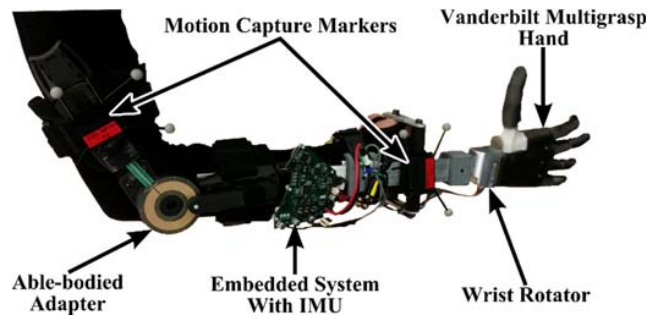


Fig. 3. Able-bodied adapter. Note that the brace immobilizes wrist rotation while allowing free elbow rotation.

elevation ($\theta_f$). In the case of a transradial prosthesis, the upper arm frontal plane angle ($\theta_a$), which is used to control wrist rotation, cannot be measured directly by the IMU, since the IMU is located in the forearm. To calculate this angle, first the gravity vector is projected onto each of the principle planes of the IMU's coordinate frame as follows:

$$\theta_x^a = atan2\left(\frac{-a_z}{a_y}\right), \theta_y^a = atan2\left(\frac{a_z}{a_x}\right), \theta_z^a = atan2\left(\frac{a_x}{a_y}\right) \quad (2)$$

where $\theta_x^a$ denotes a rotation about the x-axis (in the yz plane), as measured by the accelerometer (similarly for $\theta_y^a$ and $\theta_z^a$), and

$$\hat{g}^a = \begin{bmatrix} a_x & a_y & a_z \end{bmatrix} \quad (3)$$

is the normalized gravity vector as measured by the accelerometer. For the orientation of the IMU coordinate system shown in Fig. 1, $\theta_a$ is exactly equal to $\theta_x^a$, since the gravity vector has significant components in only the $yz$ plane if the elbow remains flexed at a right angle. In a similar fashion, $\theta_a = \theta_y^a$ if the elbow is in a fully flexed or fully extended configuration. Since the magnitude of the gravity vector in the corresponding plane gives an approximation of the contribution of that angle to the total motion, the calculation of the upper arm angle for any elbow configuration becomes:

$$\theta_a^a = \theta_x^a\sqrt{a_y^2 + a_z^2} + \theta_y^a\sqrt{a_x^2 + a_z^2} \quad (4)$$

This approximation of $\theta_a^a$ is then low-pass filtered at 3 Hz to determine a value for $\theta_a$ that is less sensitive to high frequency motions of the arm prosthesis. In this case, the angle measurement was not supplemented with gyroscope measurements, since only the relatively low-frequency angle information was of interest (i.e., angle information was intentionally limited to lower frequencies to decrease sensitivity to high-frequency movements). To calculate $\theta_f$ for the switching condition, the following formula is used:

$$\Theta_f(s) = \frac{1}{\tau s + 1}\Theta_z^a(s) + \frac{\tau s}{\tau s + 1}\frac{1}{s}\Omega^g(s) \quad (5)$$

where the capital Greek letters indicate a representation of the corresponding variables in the Laplace domain (i.e., s-domain), $\tau$ is the cutoff frequency for the low-pass and high-pass filters, and $\Omega^g$ is the angular velocity as measured

by the gyroscope. High-pass filtering the gyroscope signal and low-pass filtering the accelerometer signals in this way improves the high frequency content of the measurements and mitigates effects of linear accelerations of the arm. Both of these factors improved the ability to toggle the IMU-based controller on and off.

### F. EMG Measurement

EMG signals were obtained via two bipolar Ag/AgCl disposable electrodes (Myotronics, Inc.), attached to the anterior and posterior surfaces of the forearm in the approximate vicinity of the flexor carpi radialis and extensor carpi radialis muscles. A third electrode was attached to the elbow to serve as a ground reference for the EMG signals. The signals were differentially pre-amplified at a gain of 100, low-pass filtered at 500 Hz to remove noise, high-pass filtered at 20 Hz to remove DC offset, and rectified. The signals were then transmitted to a microcontroller (PIC32) on the embedded system and low-pass filtered at 2 Hz to provide an envelope representing the spatial integration of local EMG. The processing was performed by a custom control board which is described in more detail in [22].

Following this signal conditioning, the processed EMG were normalized based on data obtained from each subject prior to their use of the controller. The user was directed to perform flexion, extension and co-contraction actions, and these average values were used to normalize the processed signals and set the co-contraction thresholds. To detect co-contractions, a moving average of each of the EMG signals was compared to these pre-recorded co-contraction thresholds. When both signals were above their respective thresholds, the system registered a co-contraction. The velocity reference for the hand controller (in the case of the IMU-based control approach), or for both the hand and wrist controller (for the sequential approach) was calculated by taking the difference between the two EMG signals.

### G. Assessment Tasks

A Clothespin Relocation Task (CRT) and a custom task designated as the Compound Manipulation Task (CMT) were collectively used to assess the relative performance of the hand/wrist prosthesis controllers. These tasks generally involve grasping, use of the wrist, and subsequent releasing of an object, and therefore were assumed to be appropriate for assessing the relative performance of the two hand/wrist control approaches.

The CRT employs a standard graded pinch exerciser (Rolyan Graded Pinch Exerciser, Patterson Medical), which consists of a set of plastic pinch-pins (i.e., essentially clothespins), a horizontal metal rod, and a vertical metal rod, as shown in Fig. 4. Before the start of the task, three of the 9 N (2 lb) pinch-pins (the red pinch-pins in the Rolyan apparatus) were affixed to the middle horizontal rod. The test began when the subject started a timer with the contralateral hand. The subject then relocated each of the three pinch-pins to the vertical rod one at a time, then returned each to the horizontal rod one at a time, and finally stopped the timer.

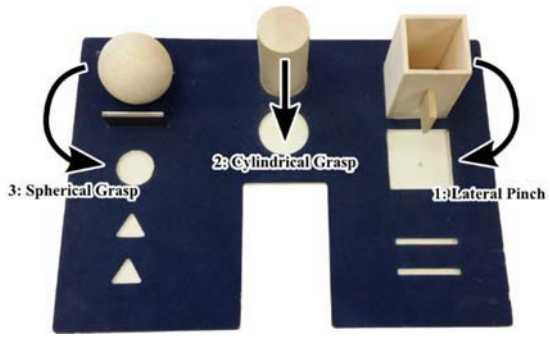


Fig. 4. Diagram illustrating the Compound Manipulation Task (CMT).

The CMT employs elements of the Southampton Assessment Procedure (SHAP), but employs these elements in a sequence that requires alternately orienting the hand in order to perform grasps. Specifically, two common hand postures during manipulation are 1) when the palm is aligned with the sagittal plane (e.g., when grasping a bottle standing on a surface), and 2) when the palm is aligned with the midsagittal plane (i.e., when grasping a bottle lying on its side). The SHAP requires both types of hand orientation, but does not require movement between the two within a timed task. The CMT essentially strings together a series of three of the SHAP assessment tasks, each of which requires an alternate palm orientation. Specifically, the first – moving an upright object using a lateral pinch grasp – requires a transverse-plane palm orientation; the second – moving an upright cylindrical object – requires a midsagittal-plane palm orientation; and the third – moving a spherical object over an obstacle – again requires a transverse-plane palm orientation. Thus, the CMT is simply a sequence of SHAP tasks, which requires alternate wrist orientations to complete. Before the start of the task, each of the objects was placed in a respective location on the far side of the SHAP tray (relative to the subject), while each subject placed his or her manipulating hand at his or her side in a neutral pose (i.e., hand hanging by the side of the body). An illustration of the initial setup is shown in Fig 4. After starting the timer with his or her contralateral hand, the subject moved each object to its corresponding location on the near side of the SHAP tray. After the last object was placed, the subject stopped the timer. The subject was instructed to manipulate each object with the type of grasp specified by the SHAP for that specific object (i.e., a lateral pinch, cylindrical, or spherical grasp). Both the CRT and CMT assessment tasks therefore characterize the ability of a subject to control wrist orientation while performing a succession of manipulation tasks, each of which generally requires a different wrist orientation. An able-bodied subject is shown performing the CRT assessment task with the experimental apparatus in Fig. 5.

### H. Able-Bodied Subject Experimental Protocol

Assessments were conducted with five able-bodied subjects (four male, one female, mean age 27 yrs). The experiments were approved by the Vanderbilt University Institutional Review Board, and each subject provided informed

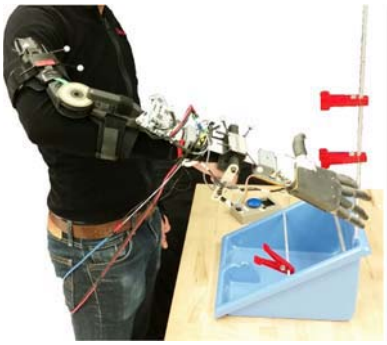

Fig. 5. Able-bodied subject performing Clothespin Relocation Test (CRT).

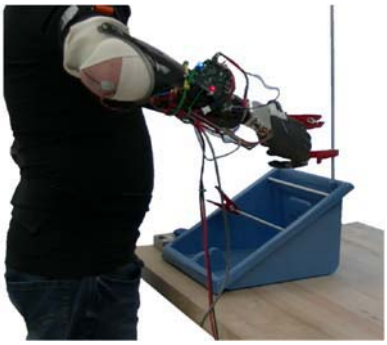

Fig. 6. Transradial amputee subject performing CRT. A video included with the supplemental material demonstrates performance of this task with both the IMU and sequential EMG controllers.

consent. Over the course of several days, the five subjects performed both the CRT and CMT tasks multiple times, first with the IMU-based controller and subsequently with the sequential EMG controller. Note that each controller was assessed on a different day, so the results should not be substantially affected by fatigue. Further, it was assumed if either controller benefitted from learning effects, the sequential controller would benefit from being assessed subsequent to the IMU controller (i.e., the order was chosen to minimize bias towards the IMU-based control approach). Note that the order could have alternatively been randomized, although given the small number of subjects, the authors felt a consistent order of presentation that minimized bias toward the IMU-based control approach was preferable.

For each controller, subjects were given approximately ten minutes to practice each task. Other than the constraints on the specific grasps to be used in the CMT (as specified by the SHAP), no instructions or constraints were given regarding performance of each task. Subjects repeated each task until task completion times fell within 10% of the mean of the three most recent trials, after which it was assumed that short-term task learning effects had effectively plateaued. Subjects generally performed each task 6-10 times before task completion times plateaued. Once the performance plateau was achieved, only the last three trials were considered for the performance assessment. For each trial, the CRT was performed first, followed by the CMT. Subjects were allowed to rest at any time, and also between tasks until they felt rested enough to continue.

### I. Assessment with a Transradial Amputee Subject

Following the controller assessments on five able-bodied subjects, the same assessment protocol was performed on a single transradial amputee subjects. The subject was a 37 year-old male, with bilateral transradial amputation. All assessments were performed with the right arm, as were the able-bodied assessments. The prosthesis prototype was mounted to the subject's daily-use socket via a standard myoelectric quick-connect. EMG signals from the two forearm sites were obtained using a pair of monopolar low-profile Ag/AgCl electrodes (Rhythmlink International), which were placed within the socket at approximately the same locations as the subject's daily-use socket electrodes. Rather than use the camera-based motion capture system employed with the able-bodied subject assessments, an IMU-based motion capture system was employed (Xsens MVN). The IMU-based motion capture system, which was not available at the time the able-bodied assessments occurred, produced the same torso movement data generated by the camera-based system, but with less apparatus. The assessment protocol employed with the amputee subject was identical to that with the able-bodied subjects, except for the fact that, due to limited availability, the amputee subject performed all testing in one day, although with a substantial rest break between the IMU and sequential EMG control cases. The amputee subject is shown performing the CRT assessment task with the prosthesis prototype and motion capture markers in Fig. 6, and a video is included with the supplemental materials showing the subject performing this task with both prosthesis controllers.

## IV. Results

The performance of the subjects for each task was quantified through self-timing and recording torso motion via the motion capture system previously described. Specifically, the motion of the torso in the coronal plane of the body (abduction of the torso) was used as the metric for assessing compensatory motion, since it is this motion that most directly correlates to rotation of the wrist. The total motion of the marker in question was represented by taking the average absolute abduction of the torso motion with respect to time. The mean of the absolute value of the signal is a convenient measure in that it is a single number that represents the amount of movement that exists in the signal relative to the neutral position of the torso, which facilitates comparisons of torso movement between tasks.

Figure 7 shows the means and standard deviations of the completion times across all subjects, separately for the group of able-bodied subjects and for the amputee subject, for the three plateaued trials corresponding to each controller, and Fig. 8 shows the corresponding average frontal plane motion. The data corresponding to these figures is given in Table II for the able-bodied subjects, and in Table III for the amputee subject. Paired $t$-tests were performed on all able-bodied subject data with 5% confidence to test for significance in results, while two-sample $t$-tests were performed on the amputee subject data, also with a 5% confidence interval. Specifically, the set of able-bodied subjects were on average

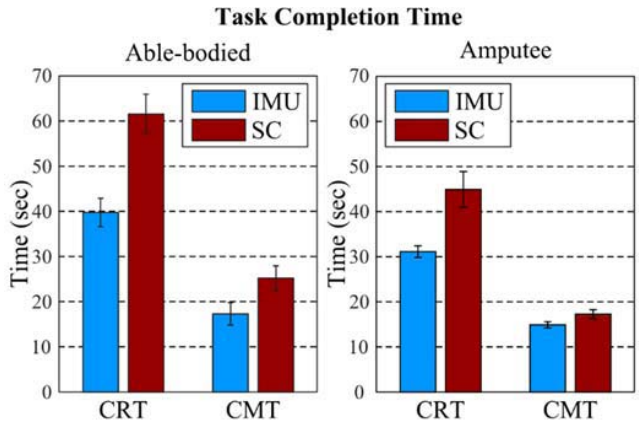


Fig. 7. Mean task completion time and standard deviations for IMU-based controller (IMU) and sequential controller (SC).

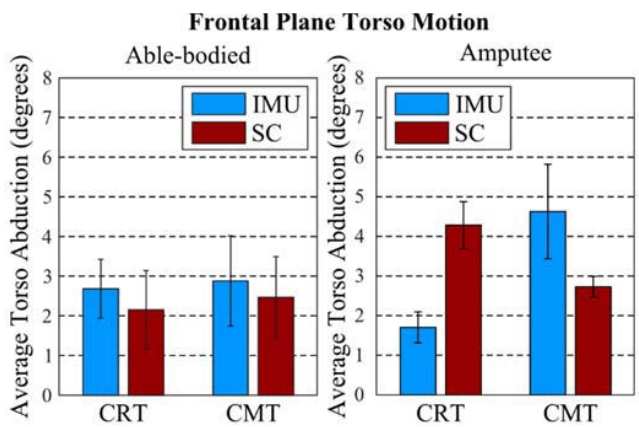


Fig. 8. Torso displacement means and standard deviations for IMU-based controller (IMU) and sequential controller (SC).

TABLE II
ABLE-BODIED PERFORMANCE DATA

| Metric | Task | Mean (STD) IMU | Mean (STD) SC |
|---|---|---|---|
| Time (s) | CRT | 39.8 (3.13) | 61.6 (4.38) |
| | CMT | 17.3 (2.5) | 25.2 (2.70) |
| Motion (deg) | CRT | 2.68 (0.741) | 2.15 (0.986) |
| | CMT | 2.88 (1.14) | 2.47 (1.02) |

able to complete the CRT in an average time of 39.8 s with the IMU controller, versus 61.6 s with the sequential EMG controller. Similarly, the subjects were on average able to complete the CMT in an average time of 17.3 s with the IMU controller, and 25.2 s with the sequential controller. The respective torso motion was on average 2.68, 2.88, 2.15, and 2.47 deg, respectively, for the IMU controller performing the CRT and CMT, and for the sequential controller performing the CRT and CMT, respectively. As such, the IMU controller enabled the subjects to perform the CMT and CRT on average 35.2% faster and 31.5% faster, respectively, than the sequential controller, with no statistically significant difference in the associated torso movement.

TABLE III
AMPUTEE PERFORMANCE DATA

| Metric | Task | Mean (STD) IMU | Mean (STD) SC |
|---|---|---|---|
| Time (s) | CRT | 31.1 (1.29) | 44.9 (3.98) |
| | CMT | 14.9 (0.683) | 17.3 (0.986) |
| Motion (deg) | CRT | 1.70 (0.391) | 4.28 (0.594) |
| | CMT | 4.63 (1.19) | 2.73 (0.263) |

The assessment results on the amputee subject followed similar trends to those of the able-bodied subjects. Specifically, the amputee subject on average (of three plateaued trials) was able to complete the CRT and CMT in 31.1 s and 14.9 s, respectively, with the IMU controller, and in 44.9 s and 17.3 s, respectively, with the sequential controller. As such, the subject was able to perform the CRT and CMT tasks on average 30.7% faster and 13.7% faster, respectively, with the IMU controller. The torso motion observed with the amputee subject showed conflicting trends; specifically, less torso motion was observed during the CRT with the IMU controller, while more torso motion was observed during the CMT with it. On average, less torso motion was observed with the IMU controller. Based on a two-sample $t$-test, the reduction in torso motion associated with the IMU controller in the CRT was significant (at a 5% confidence level), while the increase in torso motion associated with the IMU controller in the CMT was not statistically different from the motion associated with the sequential EMG controller. Thus, one could assert that the IMU controller resulted in less torso motion relative to the sequential EMG controller. Alternatively, given the inconsistent trend in means for the amputee subject, combined with the fact that the increase in torso movement associated with the IMU controller in the CMT was not statistically significant, one could more conservatively conclude that there are no clear trends associated with differences in torso motion between the two controllers, which is consistent with the able-bodied subject results.

## V. DISCUSSION

### A. Summary of Assessment Results

Assessment data indicate that able-bodied subjects were on average able to perform the CRT and CMT tasks significantly faster with the IMU-based controller as compared with the conventional sequential EMG controller, without observable differences in compensatory torso movement. The follow-up assessment with a single transradial amputee subject indicated similar performance trends in terms of both completion times and torso motion, although the performance increases were somewhat more moderate for the amputee subject, particularly when performing the CMT. Specifically, the able-bodied subjects on average performed the CRT 35.2% faster with the IMU controller, while the amputee subject performed the CRT 30.7% faster, which is well-aligned with the cohort of able-bodied subjects. However, while the able-bodied subjects performed the CMT on average 31.5% faster with the IMU controller, the amputee subject performed it only 13.7% faster (i.e., performance gains for the amputee subject in the CMT

were roughly 44% of those of the able-bodied average). Regardless, the trend of improved performance in both tasks with the IMU-based controller indicated by the able-bodied group remained prevalent in the amputee subject results. Averaging the performance improvements (i.e., reduction in completion time) between the CRT and CMT tasks, the able-bodied subjects on average were able to perform the tasks 33.3% faster with the use of the IMU controller, while the amputee subject was able to perform the tasks on average 22.2% faster.

### B. Subjective Feedback

Although subjective feedback has clear limitations, the able-bodied subjects were asked to anonymously report (i.e., via a third party) on the perceived pros and cons of each controller, and to indicate which controller they would most prefer to use on a regular basis. All subjects indicated that they preferred the IMU-based controller to the sequential EMG controller. Regarding the IMU-based controller, subjects felt the IMU controller was more intuitive and fluid, and required less task planning to use. Regarding the sequential EMG controller, subjects felt the constant use of EMG was more fatiguing, and also that the sequence of movements was less natural and required more time, relative to the IMU controller. However, subjects also expressed that the sequential controller provided better precision control of wrist orientation, and that the shoulder/wrist coupling was occasionally undesirable, for example when grasping an object close to the centerline of the body (i.e., grasping an object when the upper arm is adducted, without wanting the corresponding wrist pronation). The following are representative comments about the IMU controller from the able-bodied subjects:

- I preferred the IMU-based controller because it provided more fluid and intuitive motions. It felt much more natural and required less planning to use.
- I greatly preferred using the IMU controller as it felt more intuitive and appropriate for general tasks.
- Less effort is needed to control the wrist using the IMU wrist controller.

Some representative comments about the sequential EMG controller from the five subjects include the following:

- This controller was very easy to understand and the hand and wrist did what I wanted them to do. However, the control of this device was very slow and took a lot of planning in order to accomplish the task.
- This controller …required a lot of thinking on the user's part. From time-to-time, the co-contraction signal was hard to control, making the controller more sluggish.
- Sometimes it might be nice to have the sequential controller available for tasks involving very fine details or odd postures outside the range of the IMU coupling.
- Using the sequential controller was tiresome due to the frequent switching between hand and wrist control.

The amputee subject was also asked to report his subjective feedback (though his comments were necessarily not anonymous). He reported a generally favorable impression of the IMU-based controller, as per the following comments:

- The IMU (controller) was very easy to get familiar with.
- I like the speed of the IMU (controller).
- When I went to rotate the wrist with the arm out (abducted) it actually felt normal.

He also noted some concerns regarding the IMU controller, however, as expressed by the following comments:

- I think it (the IMU controller) would be hard to use for above-your-head tasks.
- It was sometimes hard to bring my arm so close to mid body to rotate (pronate) the wrist.

### C. Assessment Limitations

One limitation of the assessments, related to the aforementioned concern expressed by the amputee subject, is that the prosthesis was used within a relatively limited range of shoulder motion (e.g., the assessments did not require or consider overhead manipulation tasks, such as screwing a light bulb). Despite this, it's not clear how often those motions are performed, and to what extent they are performed by the prosthetic limb, relative to the type of movements incorporated in this assessment. Another limitation of the assessments, perhaps obvious, is the fact that most participants were able-bodied rather than amputee subjects; however, the fact that the amputee assessment data followed similar trends to those of the able-bodied group indicates that able-bodied results are reasonably representative of those that might be expected from a cohort of amputee subjects, although perhaps the extent of improvement may be somewhat less for amputee subjects. The latter cannot be known without performing assessments on a larger cohort of amputee subjects.

### D. Reduced Requirements for EMG

The able-bodied subjects indicated that the sequential EMG control (particularly the co-contraction command) was difficult to use consistently, or for long periods of time, due to EMG fatigue, which most likely reduced their performance in the tasks. However, this effect was not as pronounced with the amputee subject, who is more practiced with EMG control. The fact that (able-bodied) users who were not highly trained in the control of EMG experienced a more pronounced benefit with the IMU-based controller, however, may indicate that the IMU-based controller could be easier for new users to myoelectric adopt, relative to the sequential control approach. As such, the IMU control method may be especially useful for users who have poor EMG sites, or who have difficulty learning to control them. The amputee subject made a similar comment, specifically that the IMU controller would be particularly useful for patients who have poor EMG sites or trouble with EMG co-contraction.

## VI. Conclusion

In this paper, the design and assessment of an IMU-based wrist controller for a prosthetic wrist rotator was presented. The controller uses the motion of a user's upper arm as measured by an IMU to control wrist rotation. The ability of the controller to facilitate manipulation involving wrist rotation was assessed on five able-bodied subjects, and subsequently on one amputee subject, and compared to a sequential EMG

wrist controller in the performance of two tasks involving both grasping and wrist rotation. Based on the assessment, the IMU-based control approach enabled the group of able-bodied subjects to complete the tasks on average 33.3% faster than when using the sequential EMG controller. The amputee subject showed the same trend of improved performance, although on average he was able to complete the tasks 22.2% faster with the IMU-based controller (i.e., the effect size was somewhat smaller with the single amputee subject). Regardless, all subjects were able to perform the tasks significantly faster with the IMU controller, thus indicating the prospective value of supplementing EMG control input with IMU information to facilitate control of a prosthesis with myoelectric hand and wrist rotator.

In future work, the authors plan to combine the wrist controller with an IMU-based elbow controller, to facilitate control of a transhumeral prosthesis with myoelectric hand, wrist, and elbow.

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